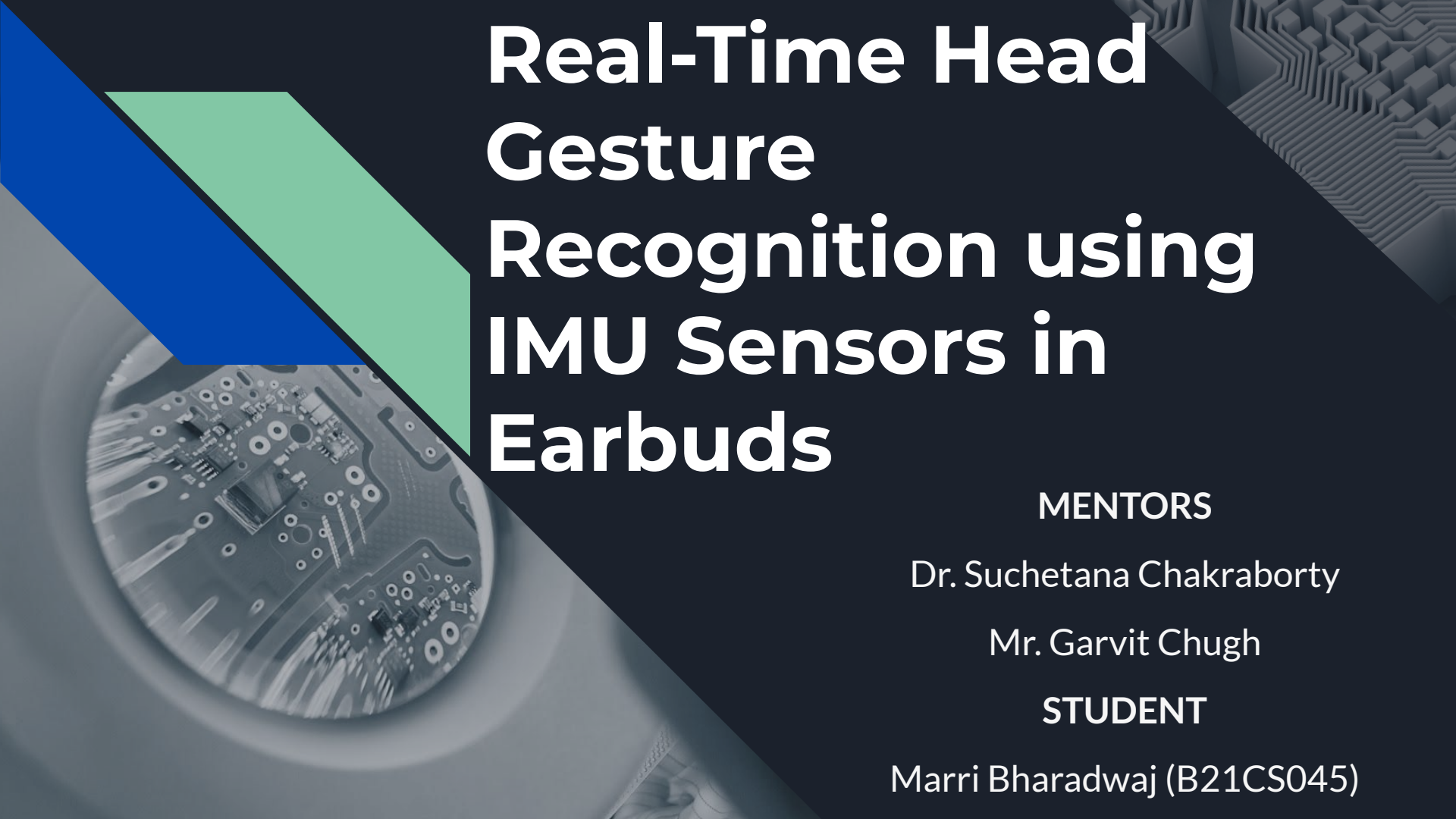




Real-Time Head Gesture Recognition using IMU Sensors in Earbuds

MENTORS

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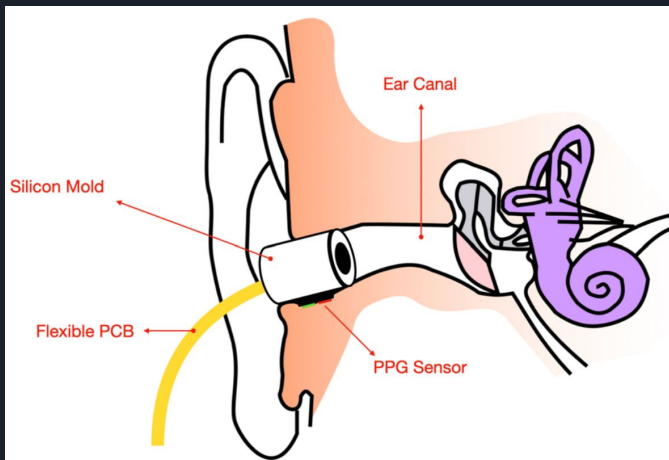
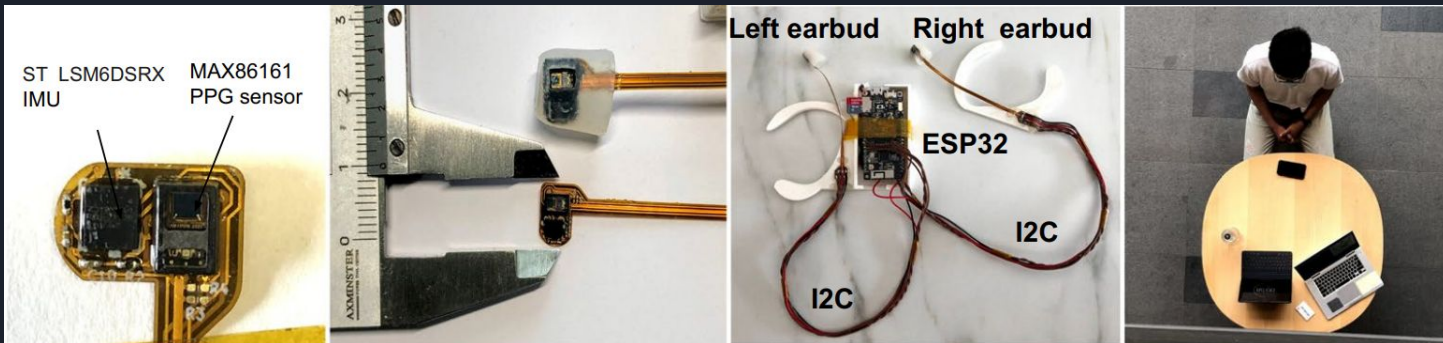
Objectives

This project focused on a multimodal dataset collected from 29 individuals performing head gestures and activities. Using IMU data (accelerometer and gyroscope) my key objectives were:

- **Feature Engineering:** Analyse statistical features to identify informative signals contributing to gesture classification.
- **Activity Classification:** Build machine learning models to recognise head gestures and facial movements using filtered and feature-engineered IMU signals.
- **Model Comparison:** Evaluate and compare multiple classifiers (LightGBM, XGBoost, Random Forest, Voting Ensemble) on accuracy, F1 score, and per-subject performance.

I aim to contribute to wearable-based Human Activity Recognition (HAR), promoting real-time, unobtrusive interfaces for accessible and context-aware computing.

Sensors



Activities

Nod	Shake	Tilt	Vertical Eye Movements	Horizontal Eye Movements	Brow Raiser	Brow Lowerer	Right Eye Wink	Left Eye Wink	Lip Puller	Chin Raiser	Mouth Stretch	Chewing	Speaking	Walking	Running
															
Head			Eyes						Mouth			head/face continuous movements		Full-body continuous movements	
One-Shot Movements												Continuous Movements			
6 mins			12 mins						6 mins			4 mins		4 mins	

Activities

Head Movements



Nod (Heads up and down)



Tilt (Left or Right)



Shake (Head left and right)

Eye Movements



Vertical Eye Movements



Horizontal Eye Movements



Left and Right Eye Wink



Brow Raiser



Brow Lowerer

Mouth Movements



Lip puller



Chin Raiser



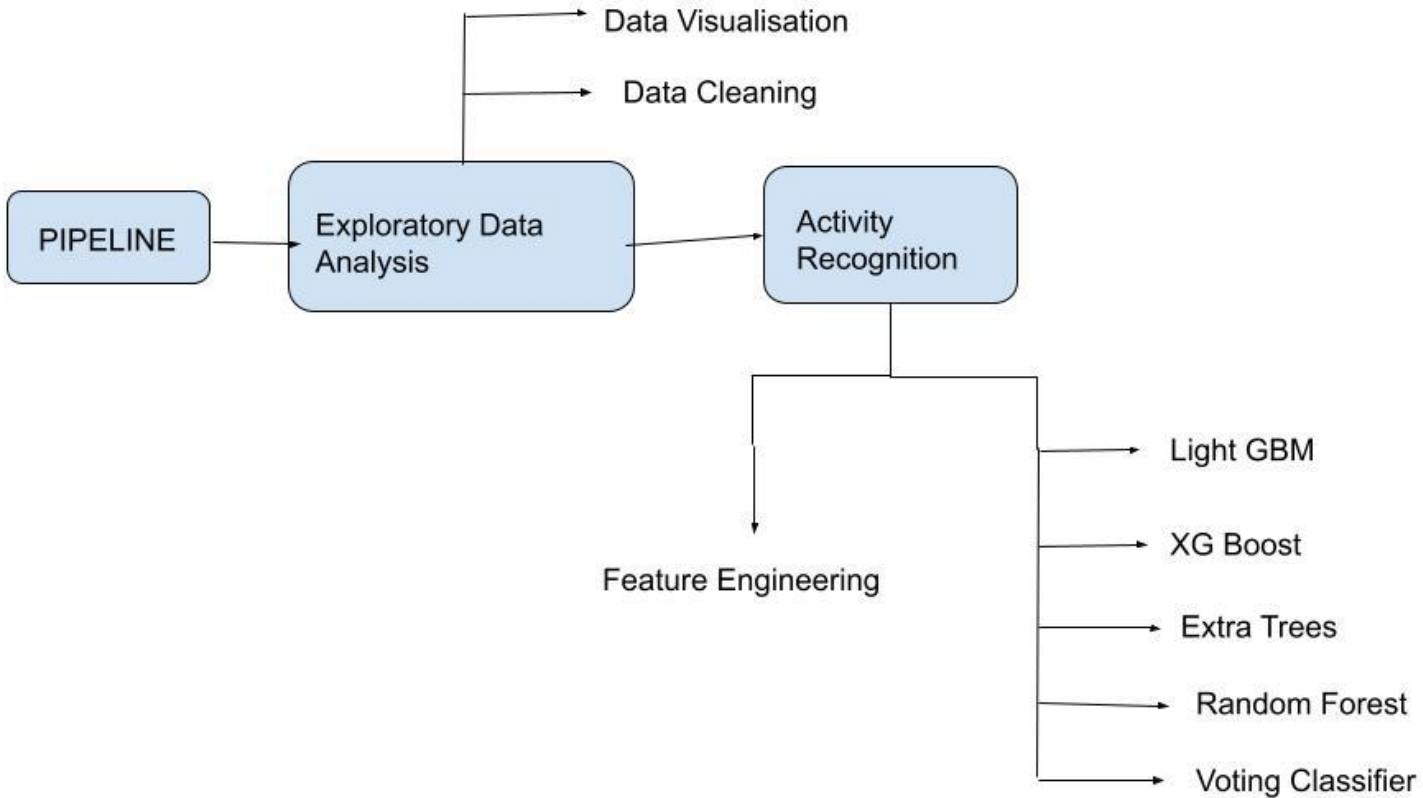
Mouth Stretch



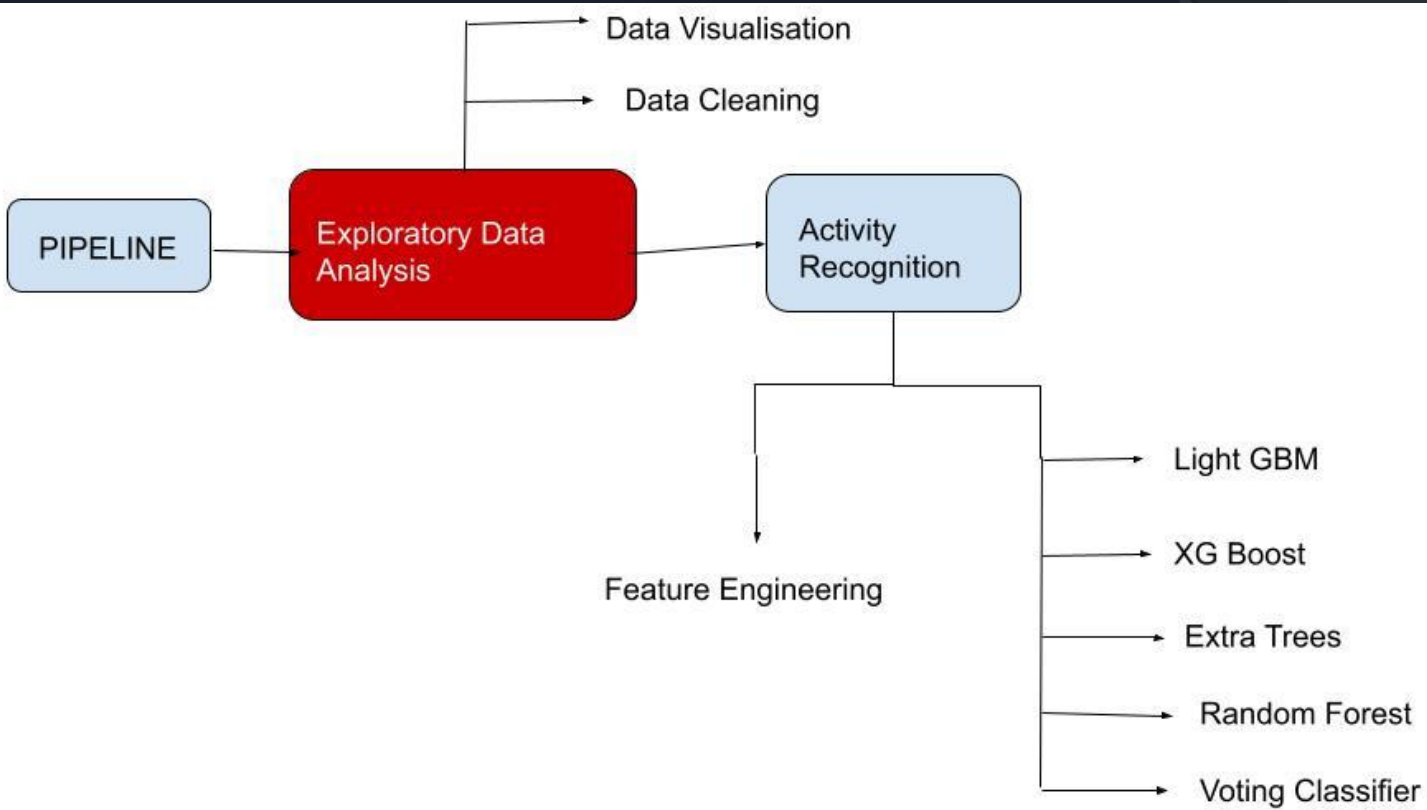
Challenges

- **Sensor Noise & Signal Drift:** Raw IMU data from earbuds included noise, requiring robust filtering (e.g., Butterworth) to extract meaningful patterns.
- **Intra-class Variability:** Same activity performed differently by users (e.g., nodding styles), leading to high variance in features across subjects.
- **Class Imbalance:** Unequal number of samples across gesture types created challenges in training balanced classifiers.
- **Real-World Dataset Size:** The size and granularity of the dataset led to longer model training and evaluation times, requiring compute-efficient solutions.
- **Subject-Dependent Patterns:** Gestures varied widely between individuals, making generalisation across subjects more difficult.

Pipeline

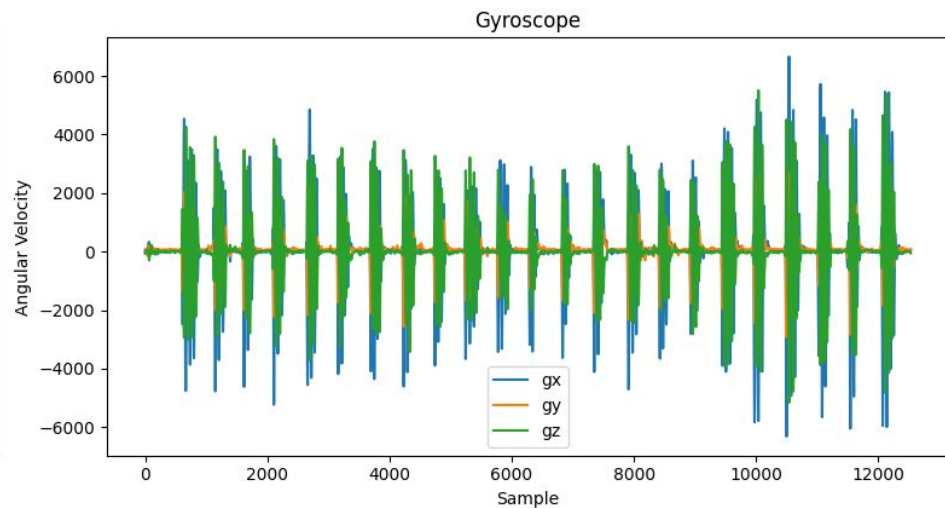
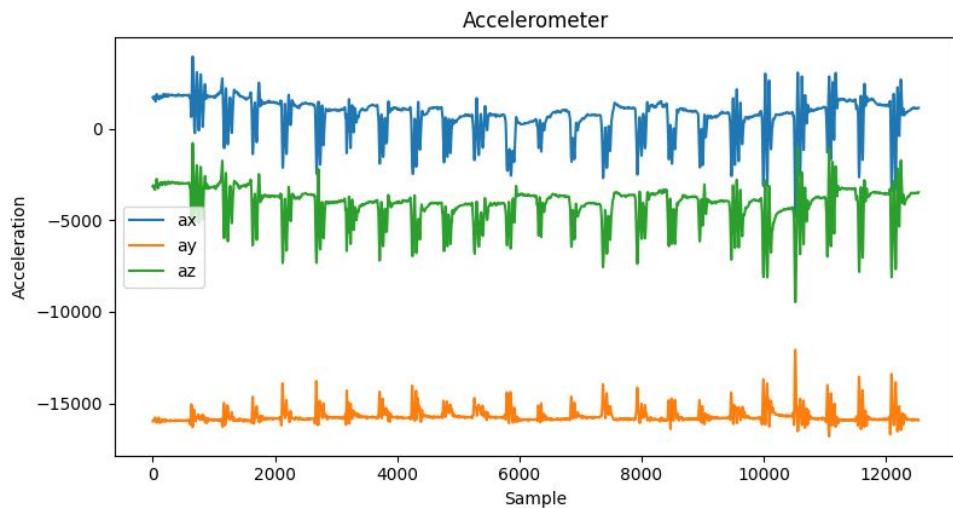


Exploratory Data Analysis (*EDA*)



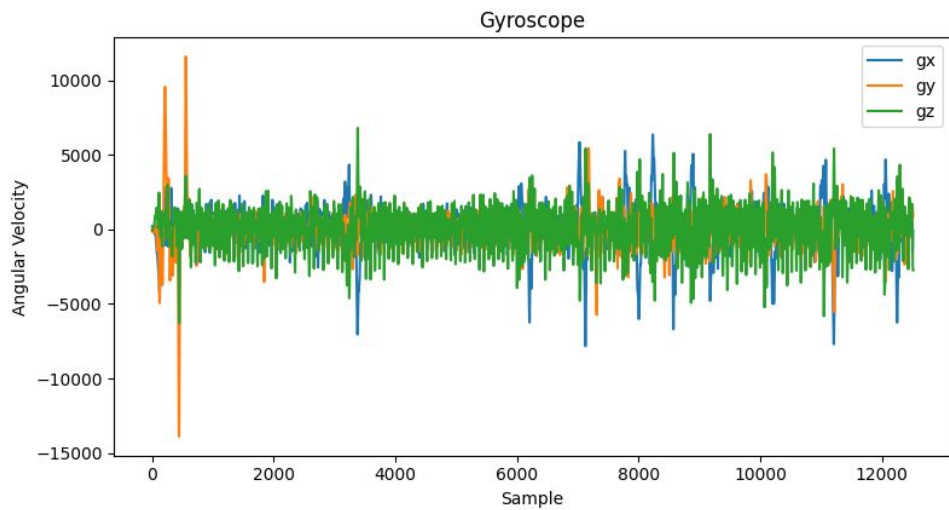
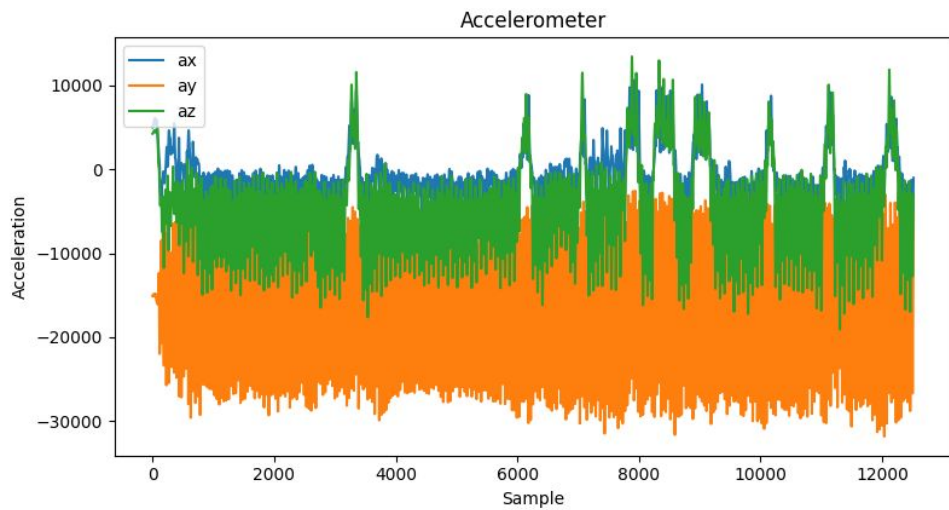
IMU Signatures

IMU Signatures: 1-nod-imu-left.csv



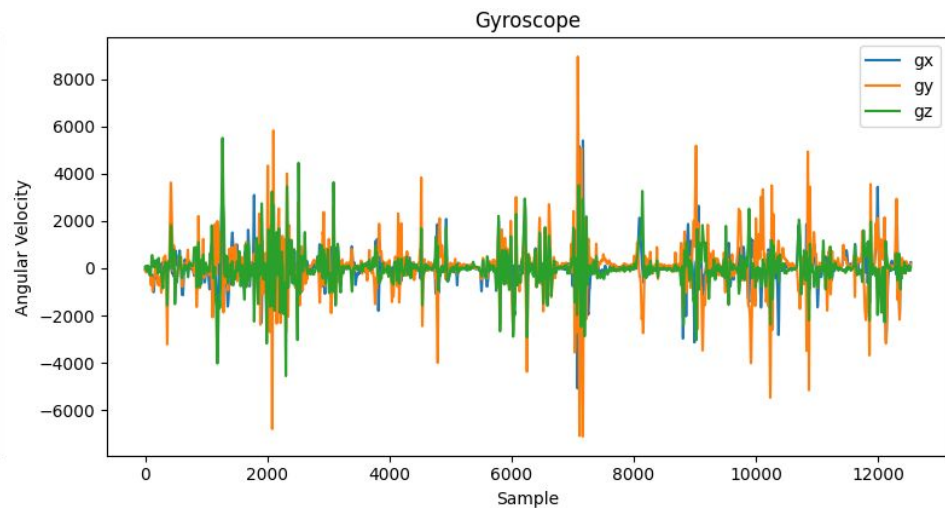
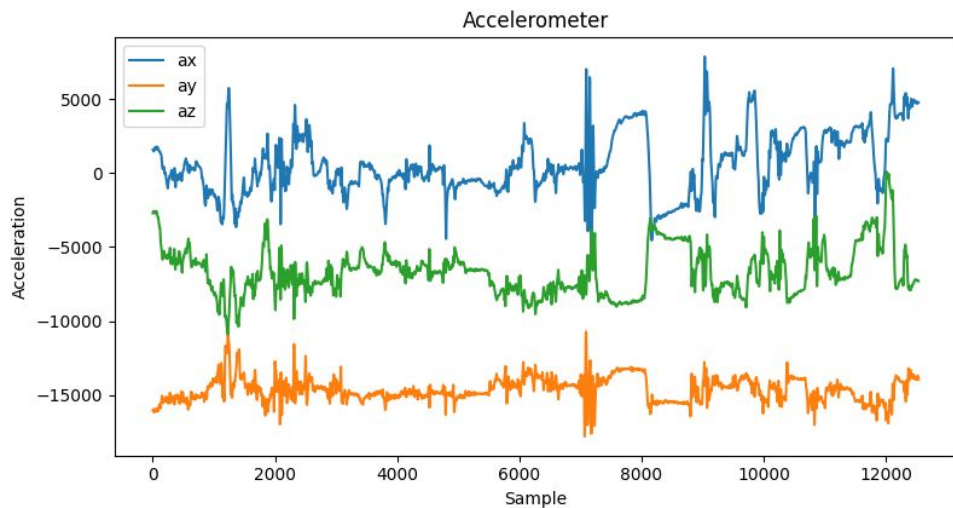
IMU Signatures

IMU Signatures: 1-running-imu-left.csv



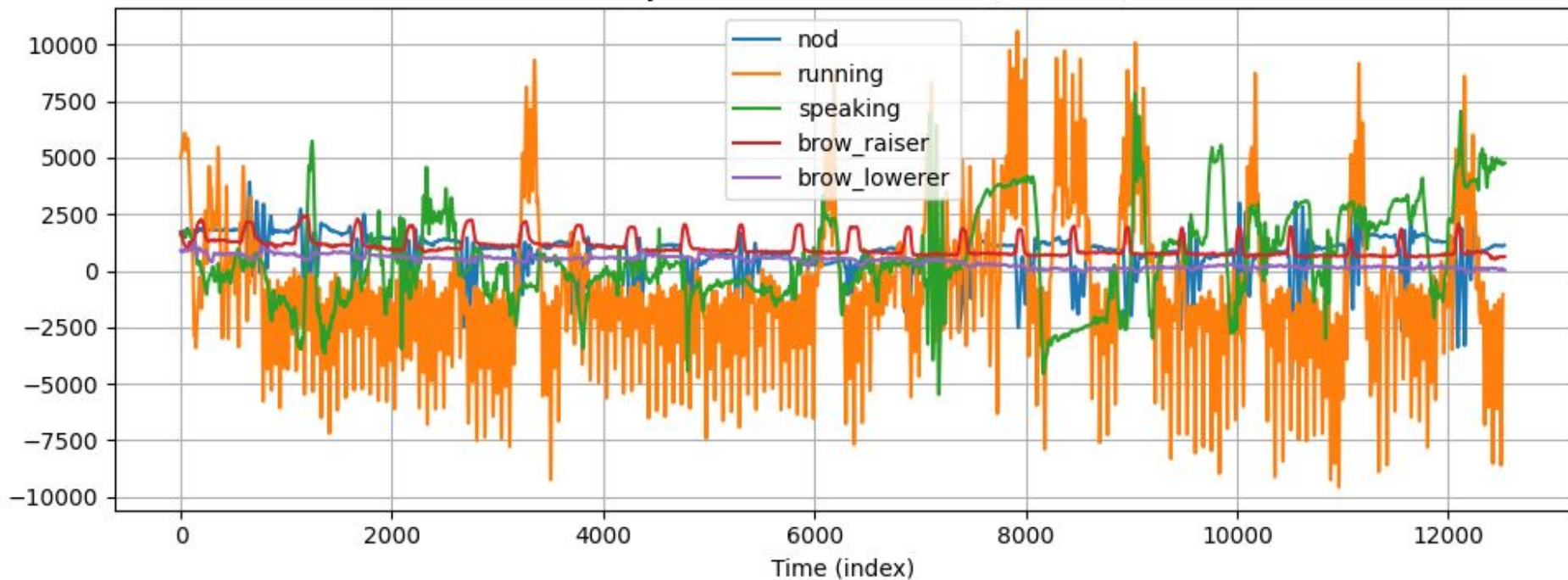
IMU Signatures

IMU Signatures: 1-speaking-imu-left.csv



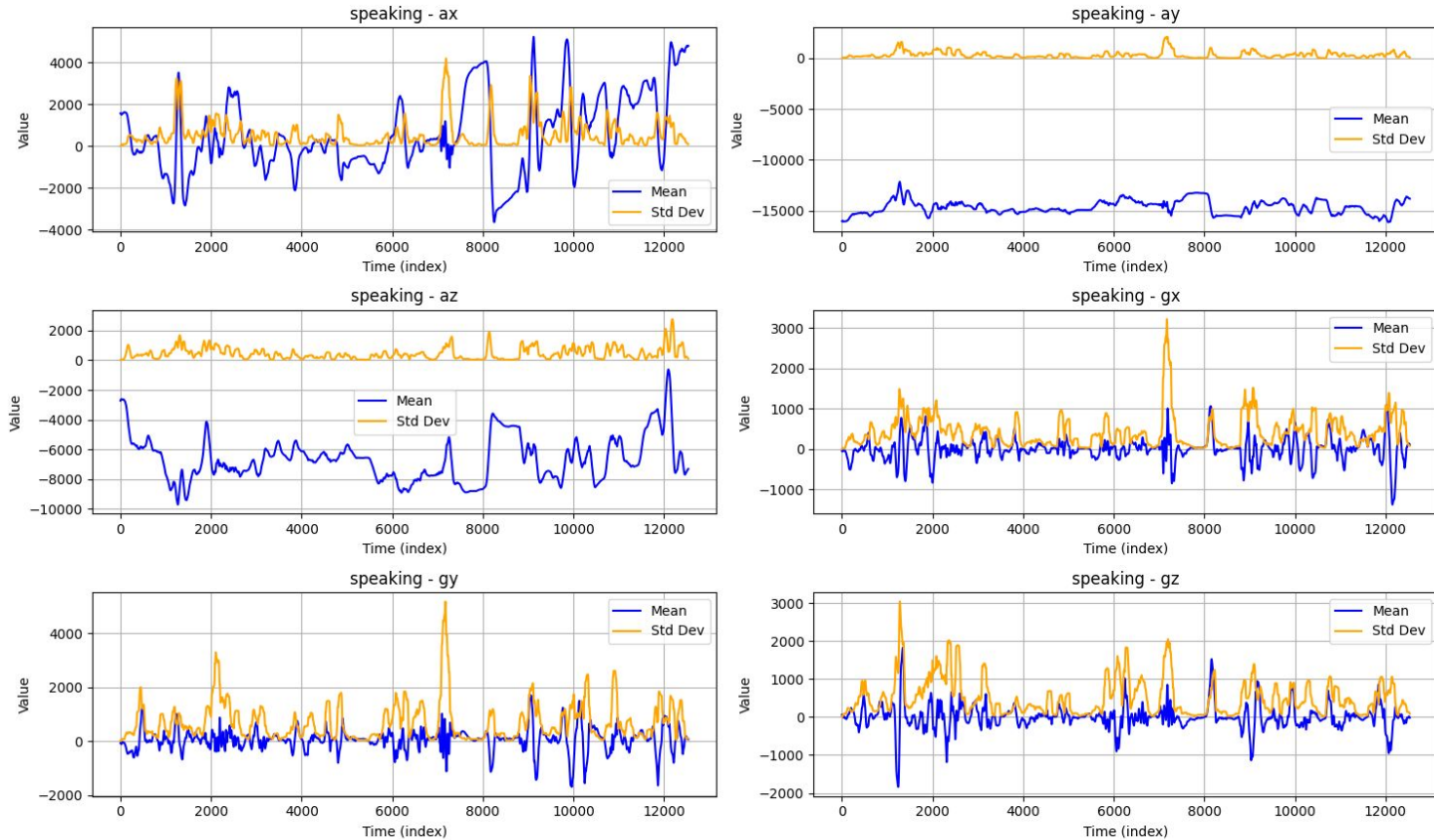
Overlay Plot

Overlay of 'ax' across activities (Person 1)



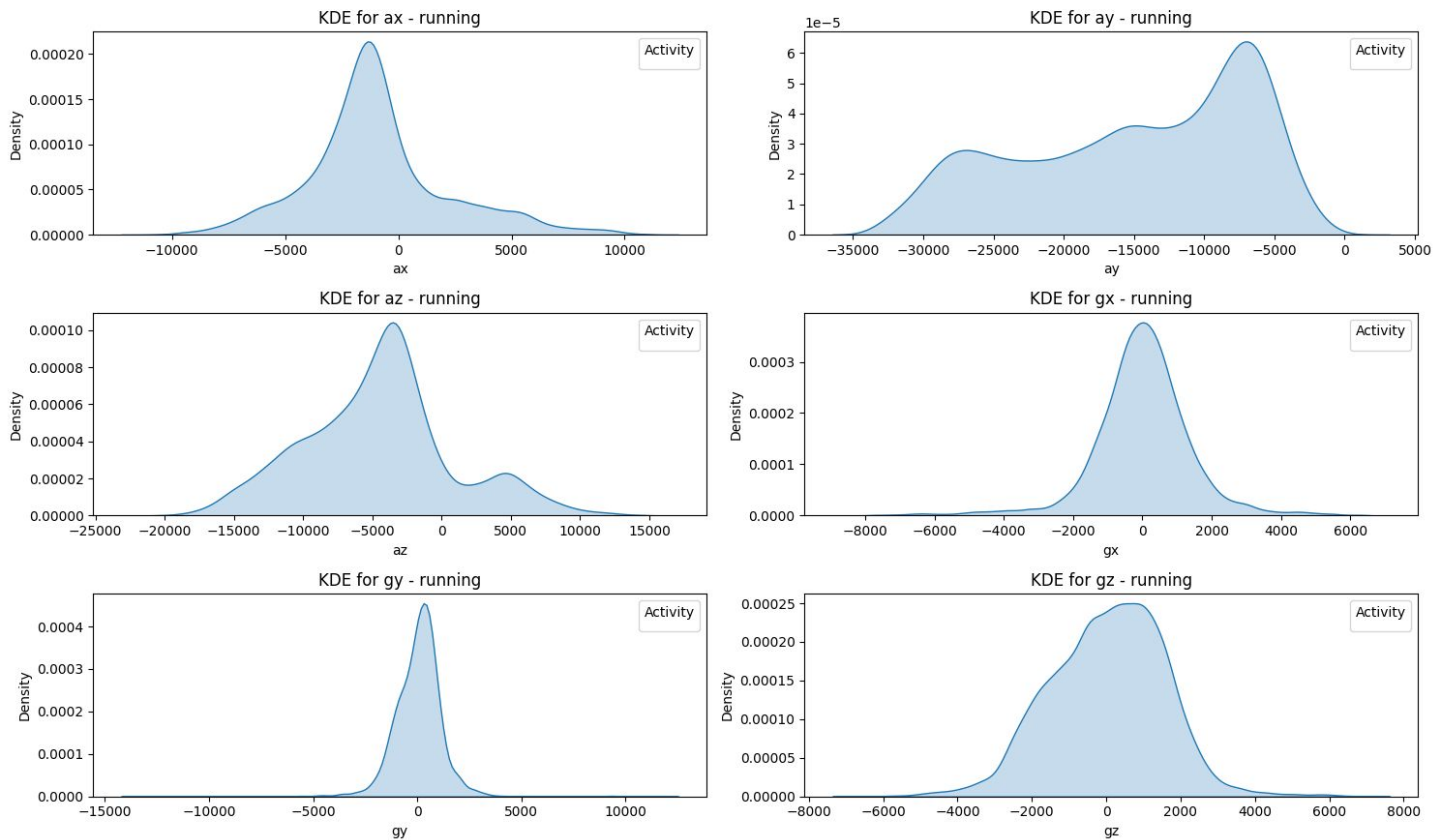
Mean/Std for Fluctuations

Rolling Mean and Std of IMU Axes - speaking

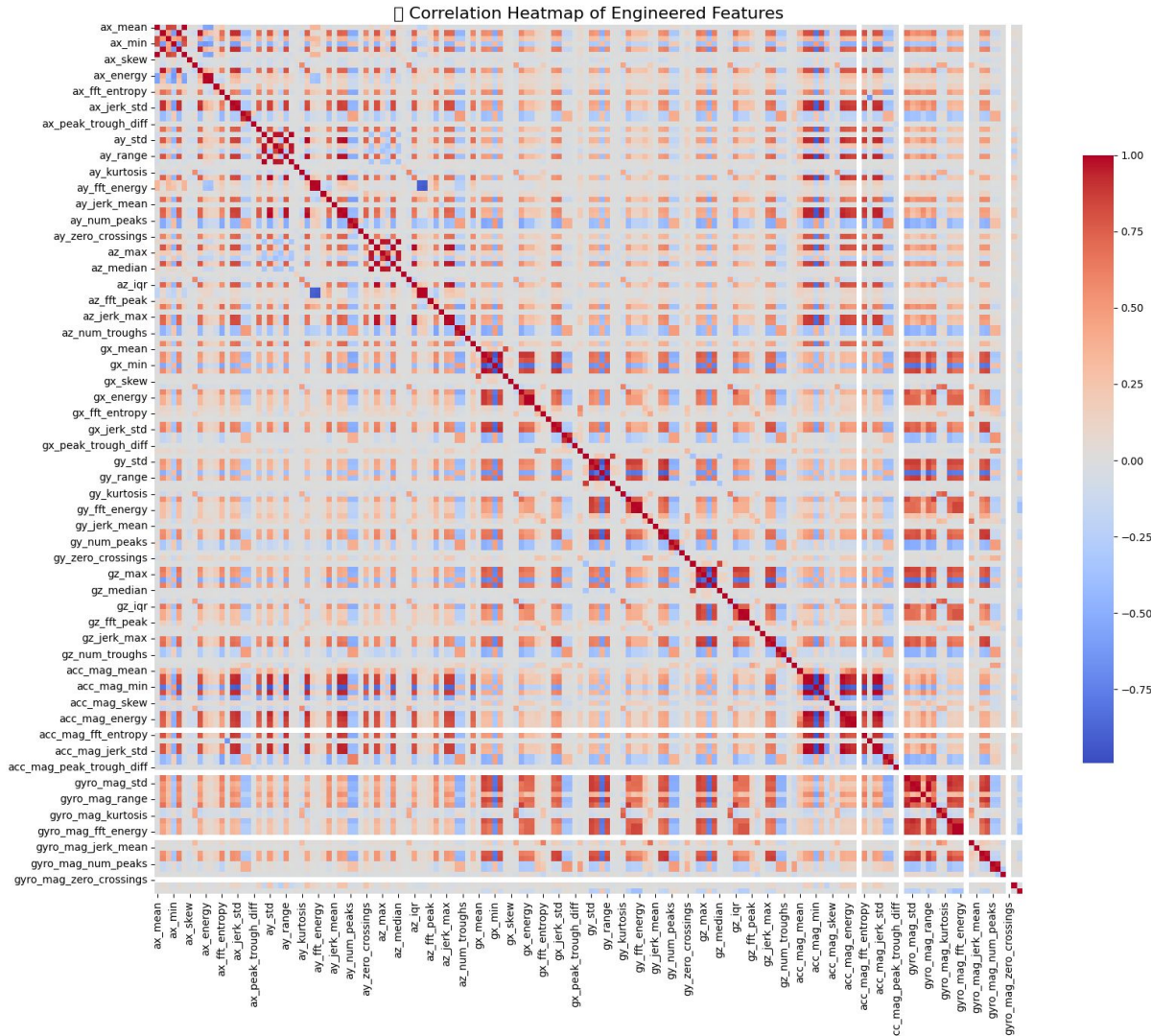


Kernel Density Function Plot

KDE Plots for IMU Axes by Activity - running

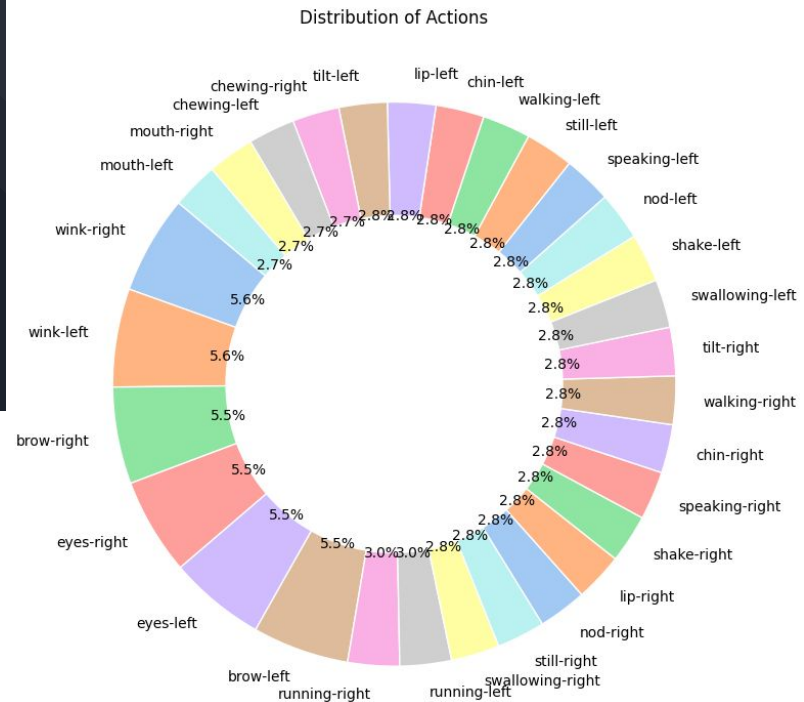
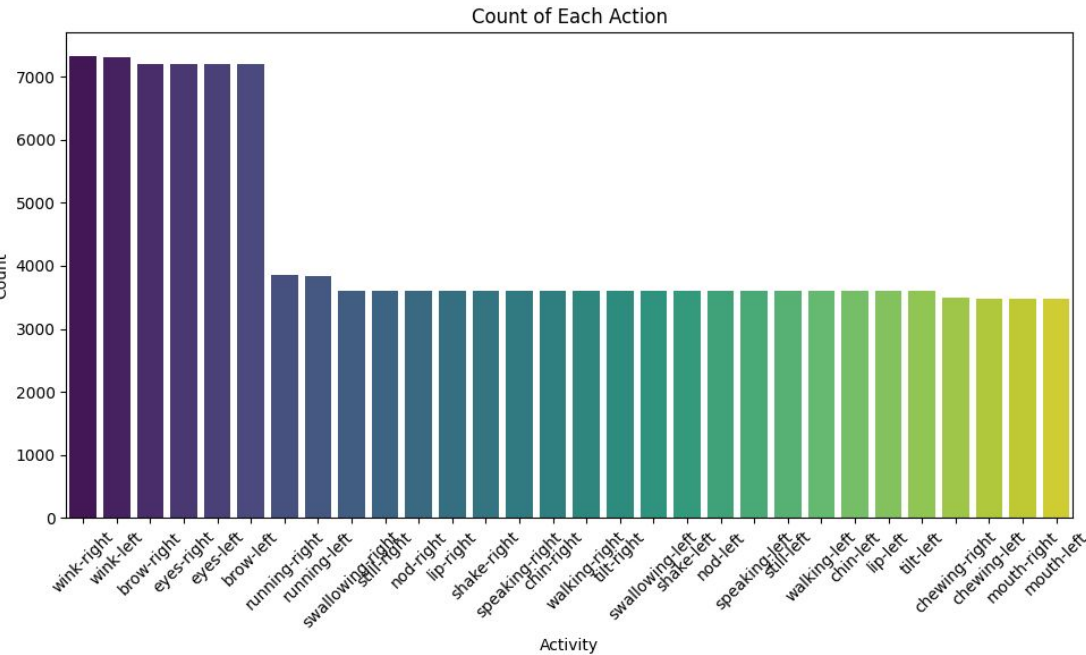


Correlation Heatmap of Engineered Features



Correlation Matrix

Distributions of Actions

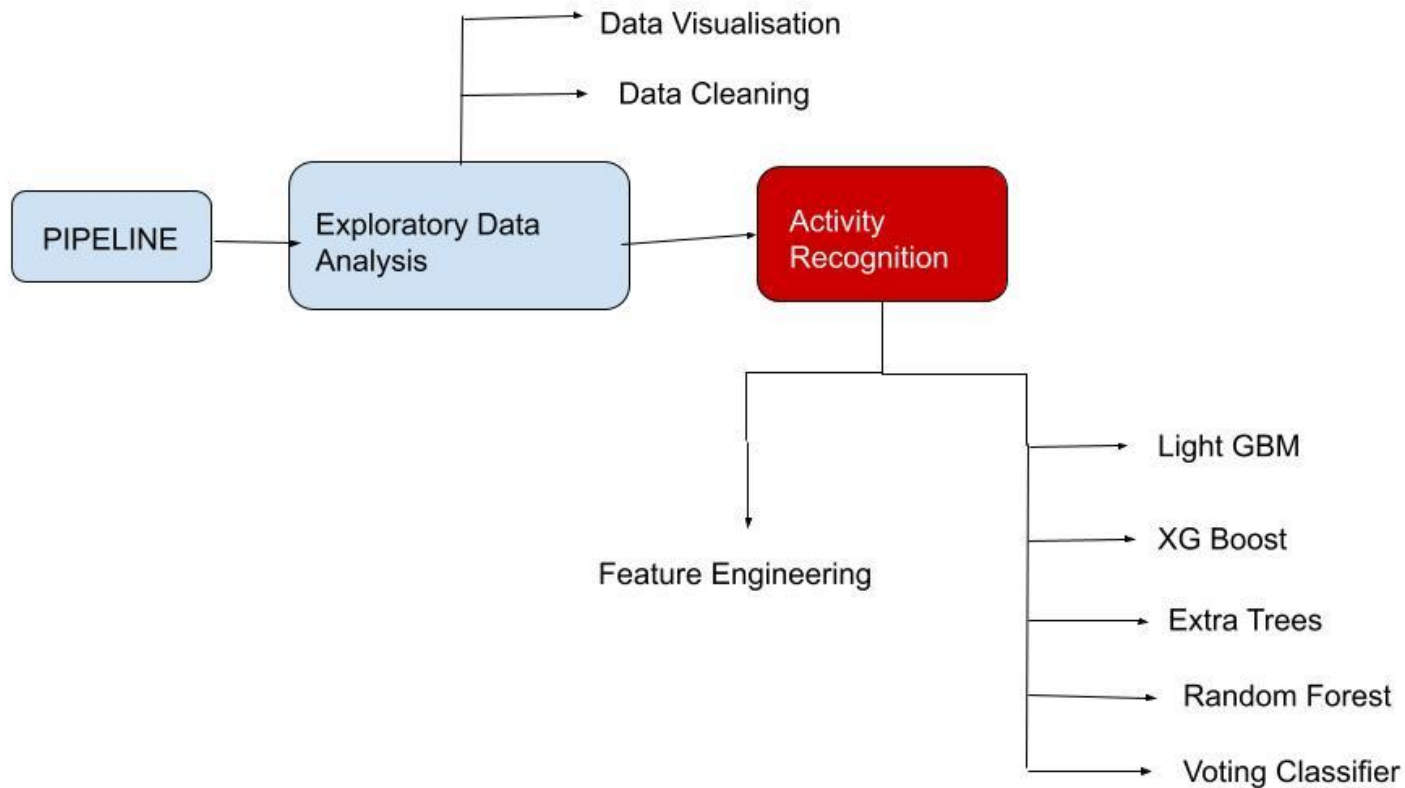




Data Cleaning & Pre-processing

- ***Timestamp Normalisation:*** Converted and aligned timestamps to maintain consistent sampling intervals across files.
- ***Sensor Signal Filtering:*** Applied a **Butterworth filter** to smooth accelerometer and gyroscope signals and reduce high-frequency noise.
- ***Missing/Corrupted Data Handling:*** Handled missing or anomalous sensor values using interpolation and statistical imputation where necessary.
- ***Label Extraction & Consistency:*** Extracted gesture labels from filenames and ensured consistent naming conventions across all samples.
- ***Participant Metadata Preservation:*** Retained person-id for later per-user evaluation and personalisation insights.

Activity Recognition





Activity Recognition

The raw IMU signals—comprising **accelerometer** and **gyroscope** readings—were segmented into fixed-length windows. For each window, I extracted a comprehensive set of over **150 engineered features**, including:

- ***Statistical Metrics***: Mean, standard deviation, max, min, skewness, and kurtosis.
- ***Temporal Features***: Zero-crossing rates, signal energy, and counts.
- ***Domain-Specific Patterns***: Relevant to movement dynamics.

The feature-labeled dataset was **label-encoded** and split using a **stratified train-test split**, ensuring class distribution balance and inclusion of person ID tracking for evaluation.

I developed and compared multiple traditional machine learning classifiers:

- **LightGBM and XGBoost**
- **Extra Trees and Random Forest**
- **A Voting Ensemble Classifier** combining the above using soft voting.

Feature Engineering

To transform raw IMU sensor readings into informative representations for classification, I applied advanced feature engineering using a combination of signal processing, statistical methods, and domain-specific insights.

FEATURE TYPE	INTUITION
<i>Time Domain (per axis)</i>	<i>Basic stats of signal in time</i>
i) Mean	Average value
ii) Standard Deviation	Variability
iii) Max, min, range	Signal bounds
iv) Median, IQR	Central Tendency and spread
v) Skew, Kurtosis	Distribution shape
vi) Energy	Signal Power
<i>Frequency Domain (per axis)</i>	<i>FFT-based signal properties</i>
i) FFT Energy	Power in frequency domain
ii) FFT Peak	Dominant frequency index
iii) FFT Entropy	Spectral Complexity

In total, there are about
162 features!

Feature Engineering

FEATURE TYPE	INTUITION
<i>Jerk Stats (per axis)</i>	<i>First derivative motion features</i>
i) Jerk Mean	Average acceleration rate
ii) Jerk Max, Standard deviation	Dynamics of change
<i>Peaks/Trough (per axis)</i>	<i>Local extrema analysis</i>
i) Num peaks	High points in signal
ii) Num troughs	Low points
ii) Peak-trough diff	Balance of movement
<i>Zero-crossings (per axis)</i>	<i>Signal direction changes</i>
<i>Vector Magnitude Features</i>	<i>Combined xyz motion (acc & gyro)</i>
<i>Cross-correlations</i>	<i>Signal sync between axes</i>
Corr(ax, ay), Corr(gy, gz)	Correlated movement axes



Intuition-Windowed Segmentation



By dividing continuous accelerometer and gyroscope signals into overlapping windows (4-second windows with 2-second overlap), I created structured segments that capture temporal dynamics while maintaining context between adjacent frames. This approach allows for:

- Consistent processing of variable-length recordings
- Enhanced signal analysis through overlapping segments that reduce boundary effects
- Improved model generalisation by creating multiple training examples from each recording

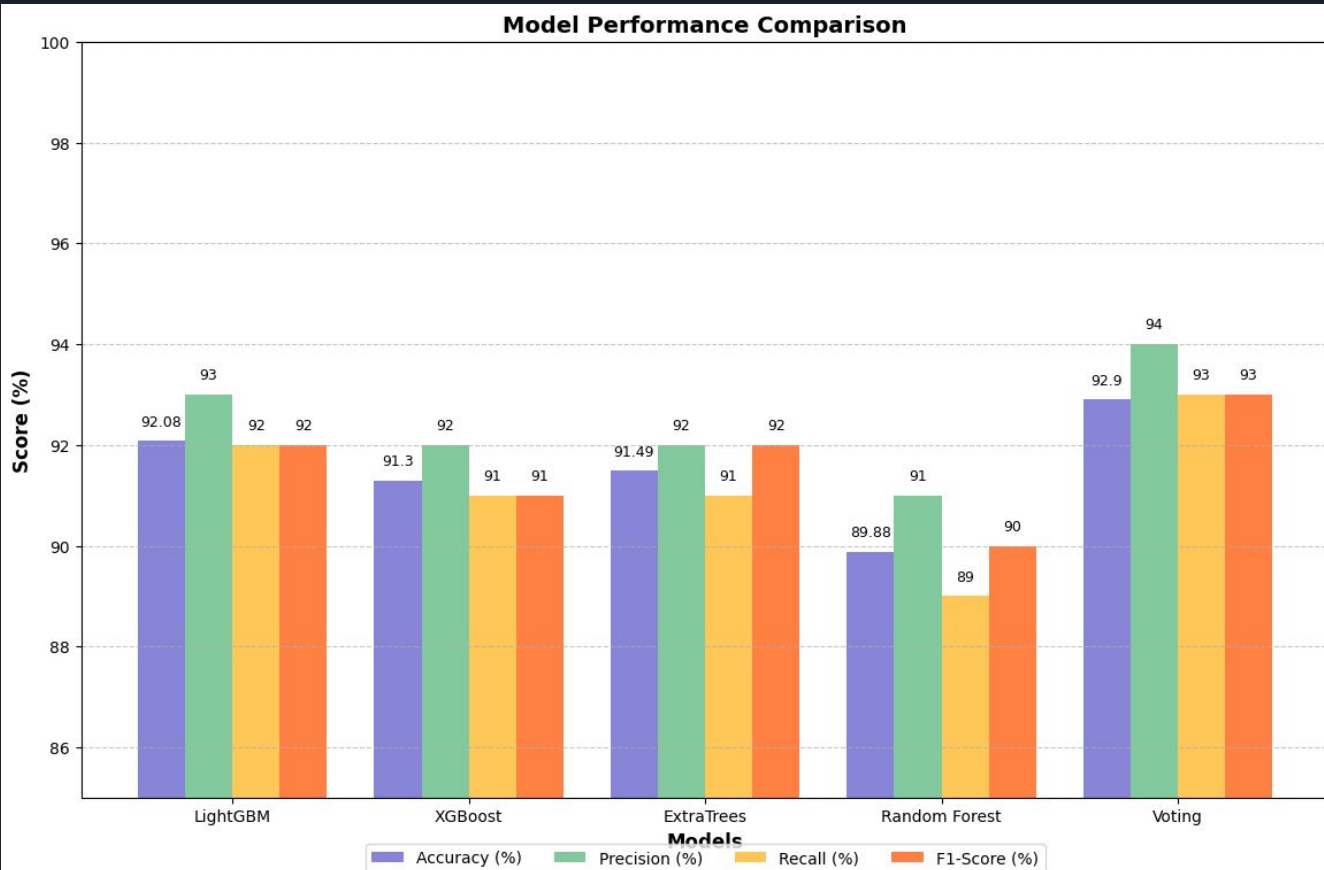
My implementation carefully handles data quality by skipping windows containing NaN values or producing extraction errors. This preprocessing pipeline significantly improved model performance compared to non-windowed approaches, enabling my classifier to better recognise subtle facial expressions from earbud IMU data.

Models Comparison

→ The Voting Classifier clearly outperforms all individual models with the highest accuracy (92.90%).

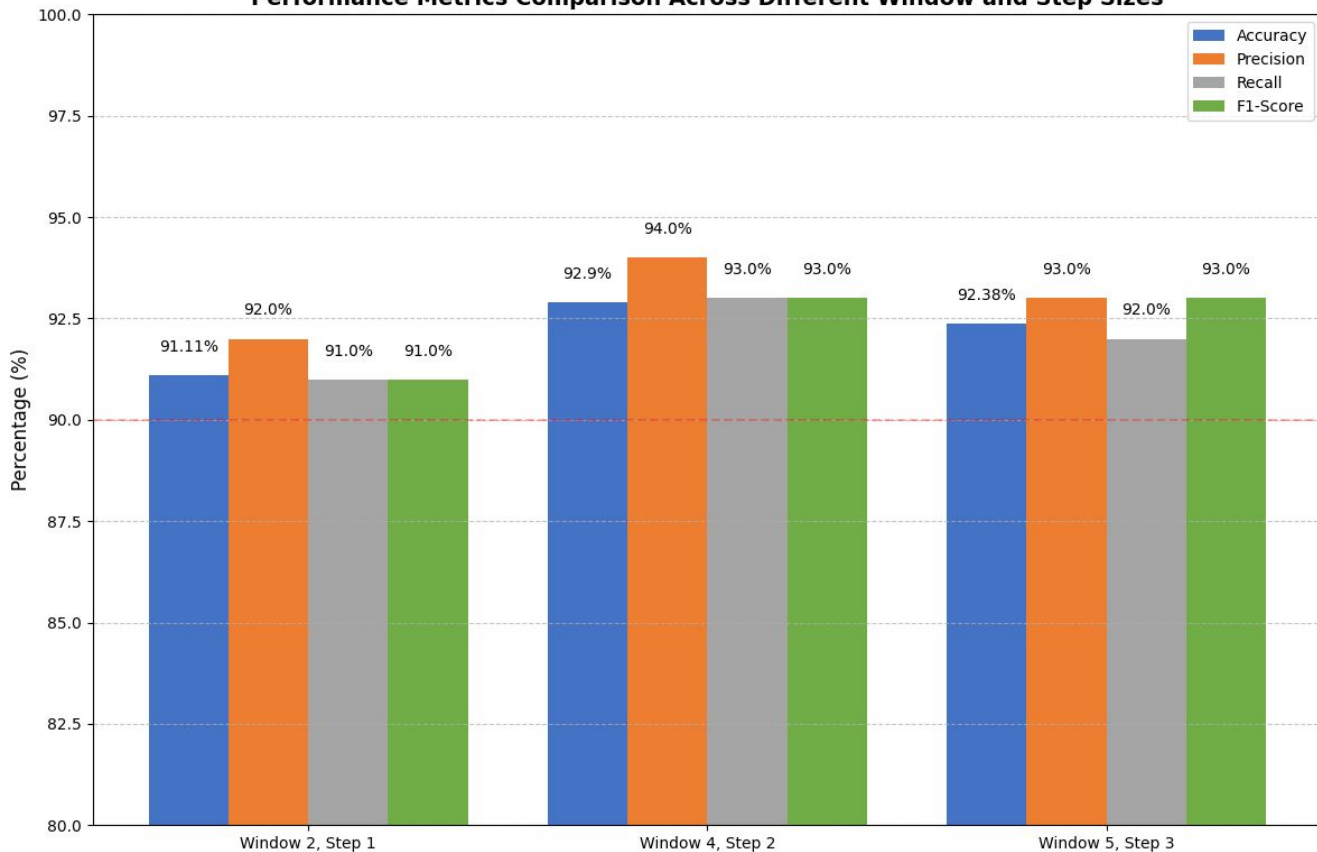
→ LightGBM follows next with 92.08% accuracy, while Random Forest lags behind at 89.88%.

→ The tree-based boosting models (LightGBM and XGBoost) show strong performance, though the margin between the top four models is relatively small (within ~1% accuracy).



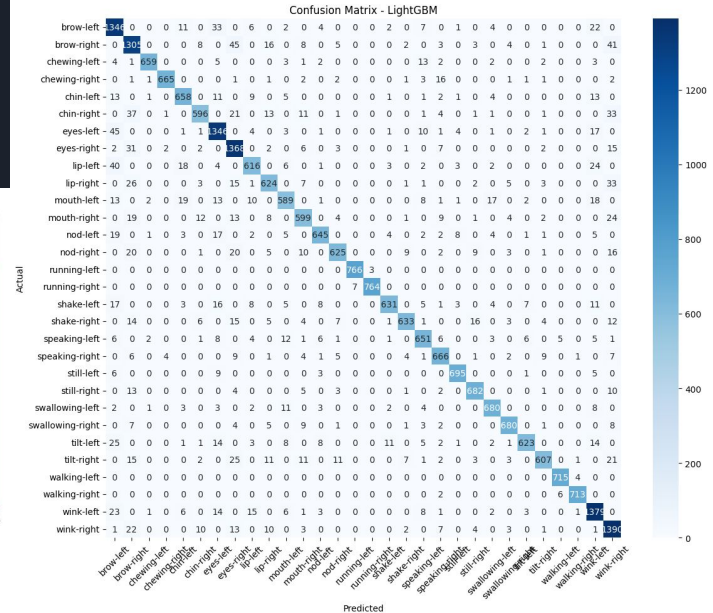
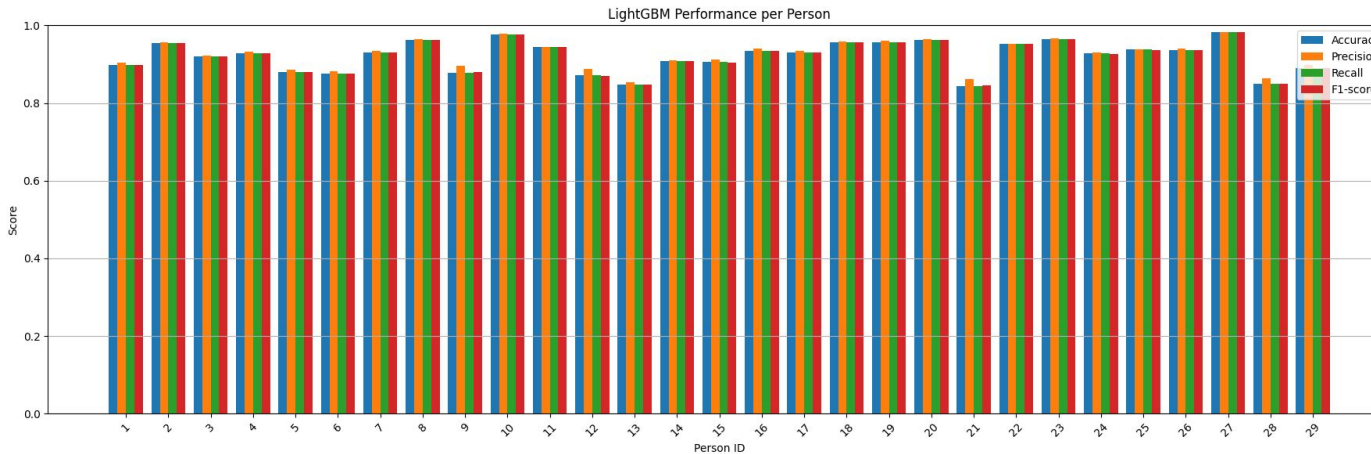
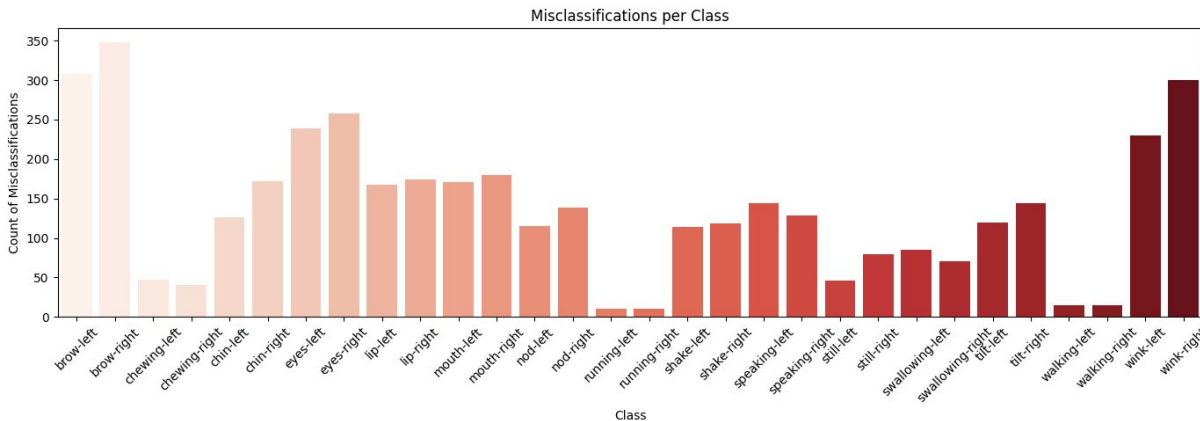
Hyper-parameter Tuning

Performance Metrics Comparison Across Different Window and Step Sizes



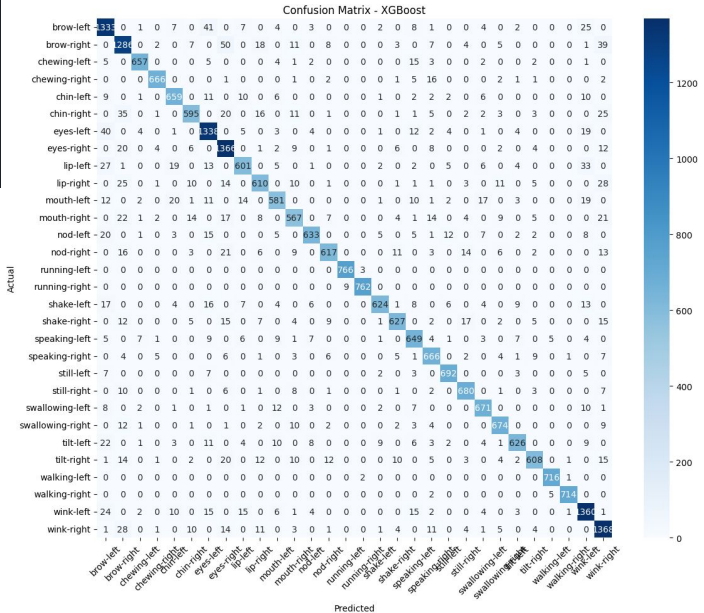
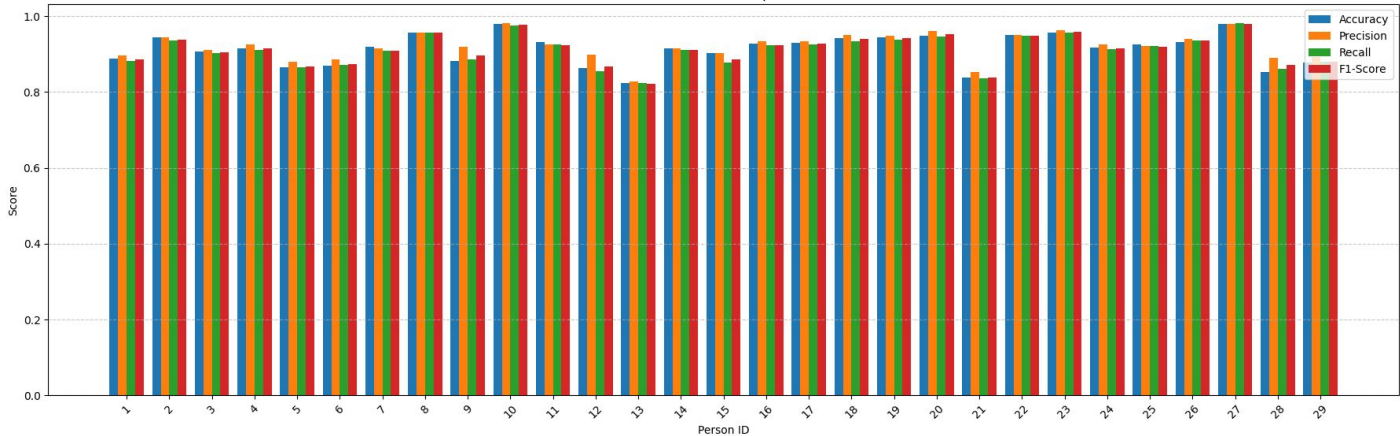
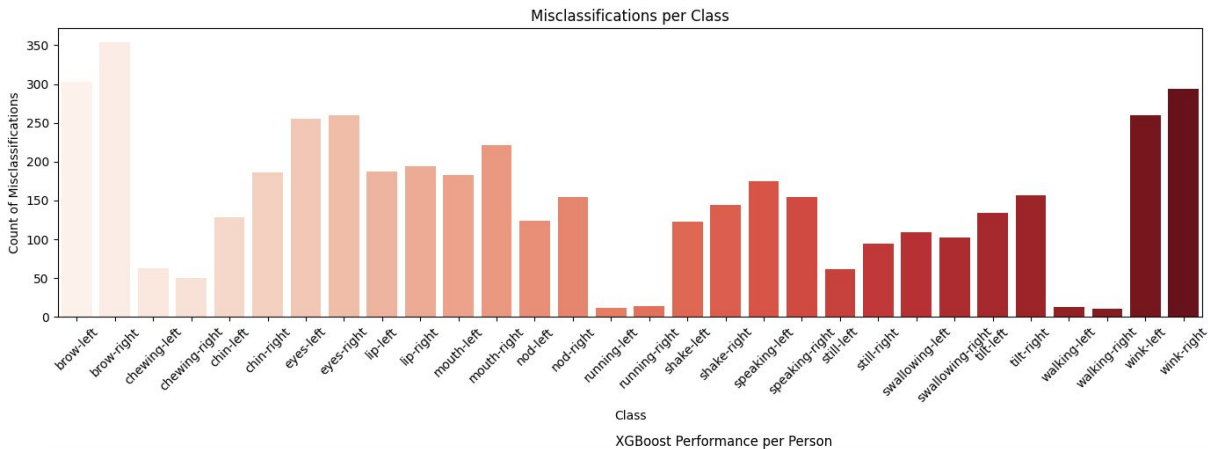
I varied the window size and step size to check whether zooming in or zooming out on the timeframe would prove useful. Surprisingly, my models performed the best on optimum time frame of window size as 4 sec and step size as 2 sec; when compared to zooming out (windows sizes as 5 sec; step size as 3 sec) or zooming in (windows sizes as 2 sec; step size as 1 sec).

Best parameters of (4 sec, 2 sec)



LightGBM

Best parameters of (4 sec, 2 sec)

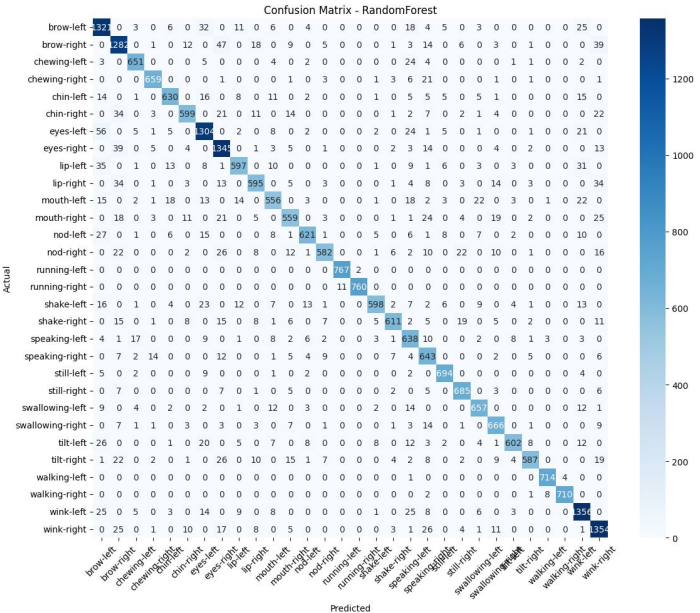
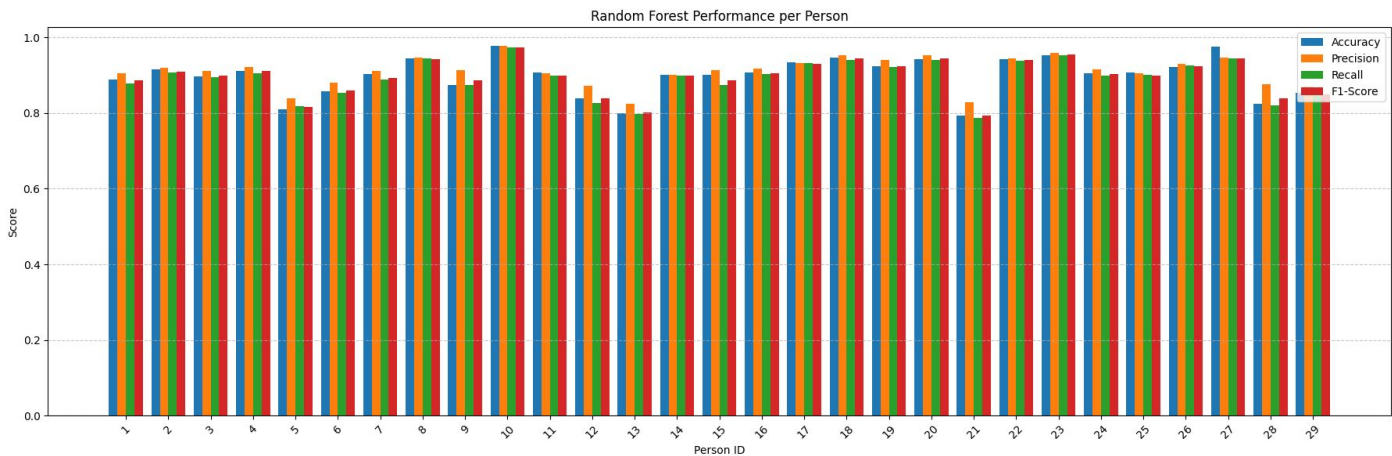
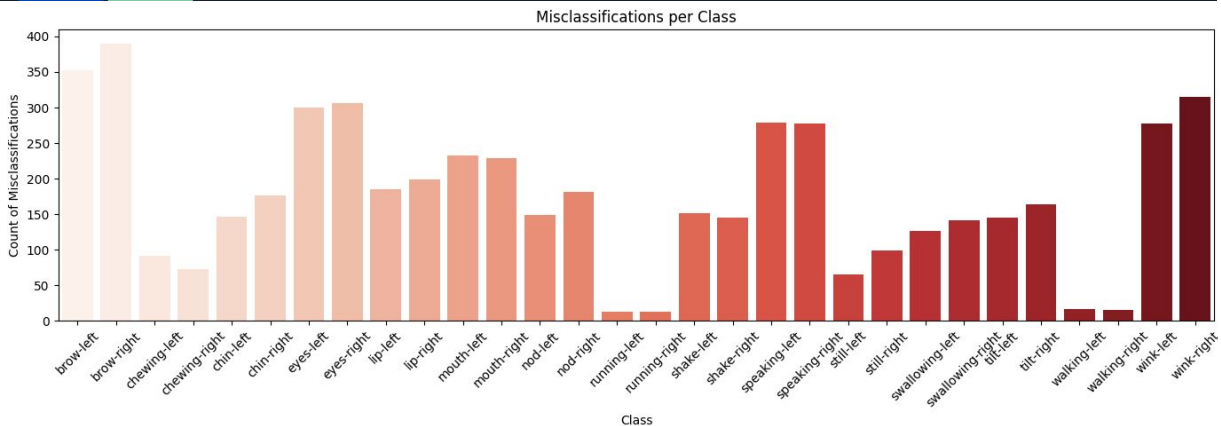


XG Boost



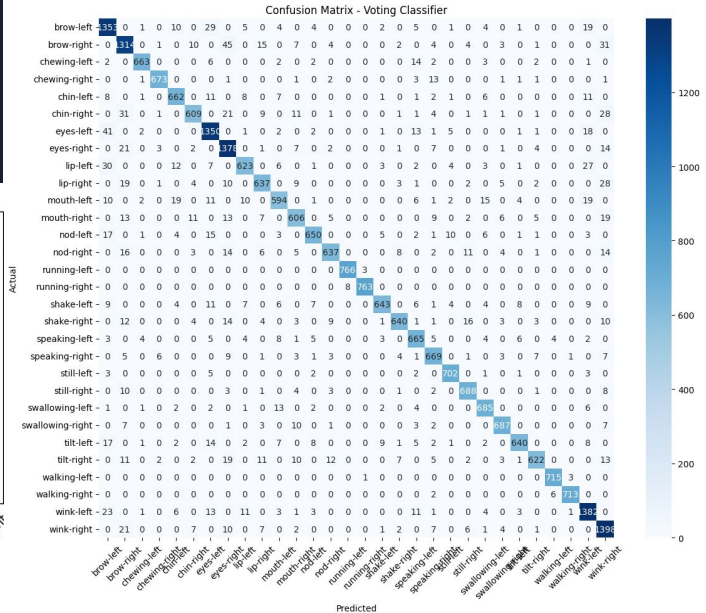
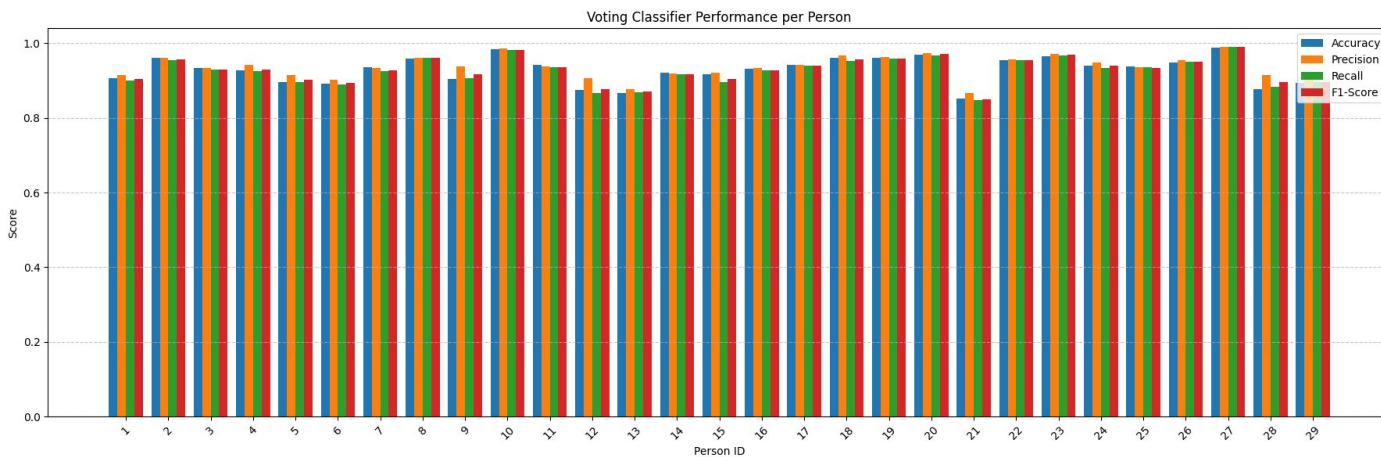
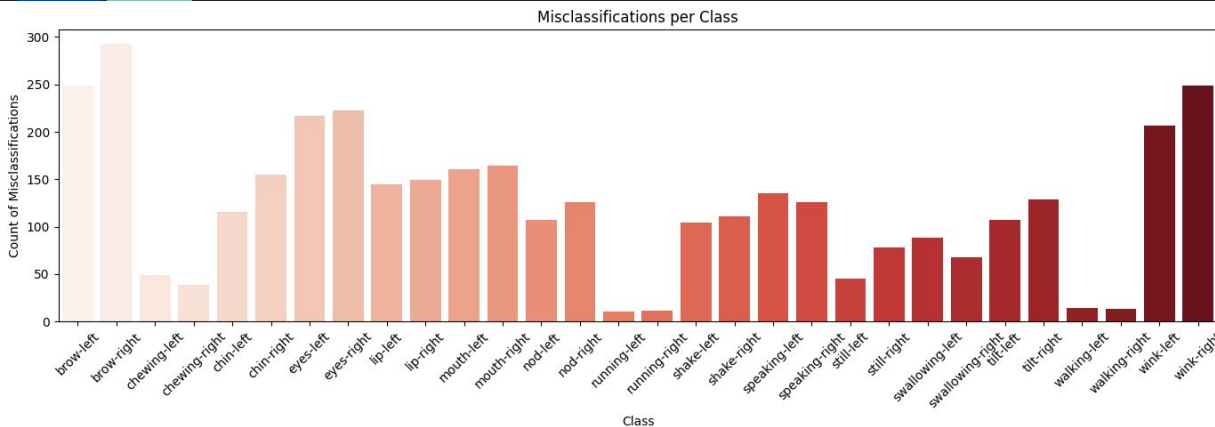
Extra Trees

Best parameters of (4 sec, 2 sec)



Random Forest

Best parameters of (4 sec, 2 sec)



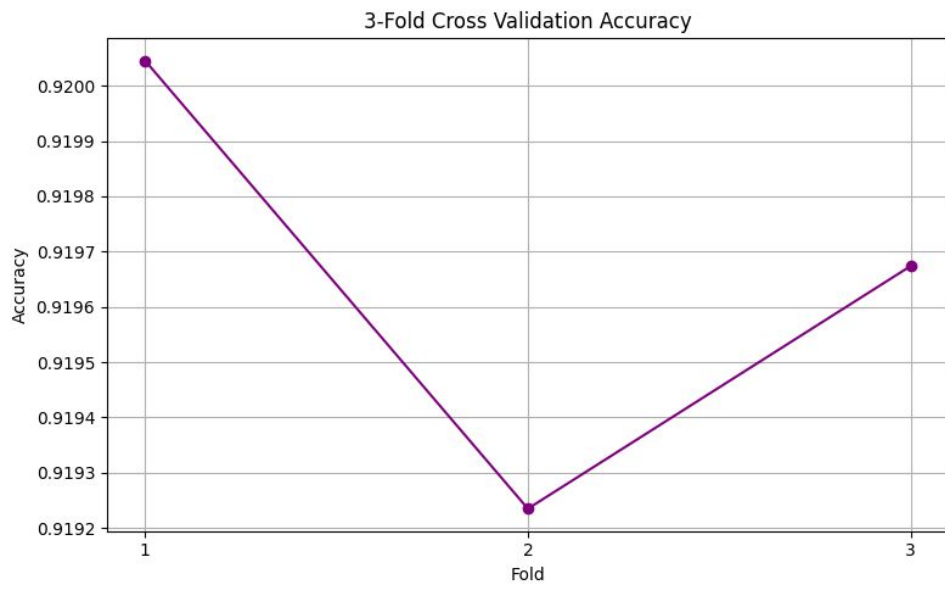
Voting Classifier

K-fold Validation

Implementing 3-fold stratified cross-validation provided rigorous validation. I created three balanced folds while preserving class proportions, ensuring each action is properly represented across training and testing sets.

This approach confirmed my Voting Classifier's reliability with consistent cross-fold accuracy and low standard deviation, regardless of data partitioning.

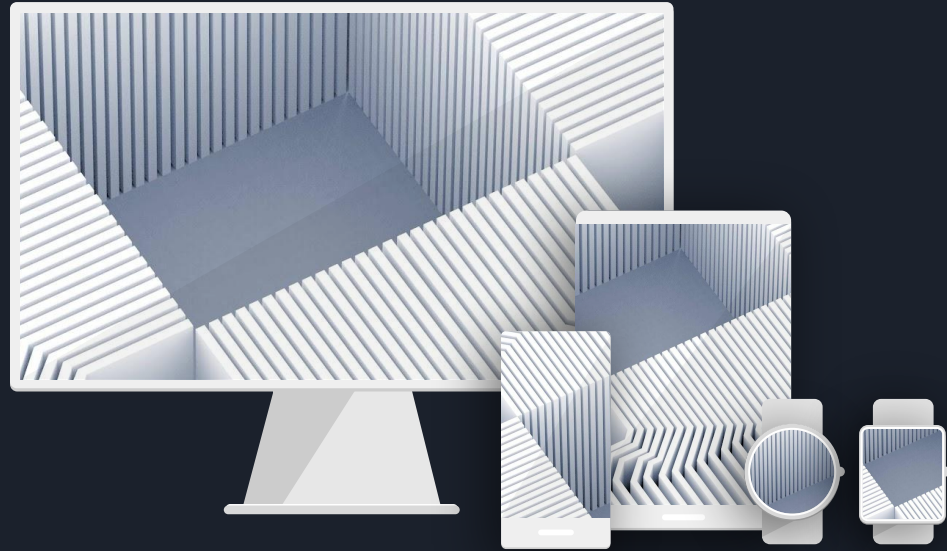
- **3-Fold CV Accuracies:** [0.92004528 0.91923486 0.9196738]
- **Mean Accuracy:** 91.97 %
- **Standard Deviation:** 0.0003



The exceptionally low standard deviation (0.0003) confirms my model's stability across different data partitions, while the high mean accuracy (91.97%) validates the effectiveness of my approach.



References



1. [Andrea Ferlini, Alessandro Montanari, Ananta Narayanan Balaji, Cecilia Mascolo, & Fahim Kawsar. \(2023\). EarSet: A Multi-Modal In-Ear Dataset \[Data set\]. Zenodo. <https://doi.org/10.5281/zenodo.8142332>](https://doi.org/10.5281/zenodo.8142332)
2. [Vimal Mollyn, Riku Arakawa, Mayank Goel, Chris Harrison, and Karan Ahuja. 2023. IMUPoser: Full-Body Pose Estimation using IMUs in Phones, Watches, and Earbuds. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems \(CHI '23\). Association for Computing Machinery, New York, NY, USA, Article 529, 1–12. <https://doi.org/10.1145/3544548.3581392>](https://doi.org/10.1145/3544548.3581392)